

ORF 524 — Midterm Exam

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Notes:

- Duration of examination: 180 minutes.
An additional 15 minutes are given to upload your solutions to Gradescope.
- All parts are equally weighted (5 points).
Please provide as much detail as possible in your answers.
Answers without proper justification will not receive full credit.
- Please start each question on a separate page.
A single PDF file must be submitted to Gradescope.
- Only one page of personal notes are allowed during the examination.
This is otherwise a closed-book and individual examination.

1 Finite-Sample Statistical Inference (50 points)

Let $(X_i : 1 \leq i \leq n)$ be an i.i.d. random sample from $X \sim \mathbb{P}_\theta$, with absolutely continuous d.f.

$$F(x; \theta) = \mathbb{P}_\theta(X \leq x) = \left(1 - \left(\frac{L}{x}\right)^{1/\theta}\right) \mathbb{1}(x > L),$$

where $\theta \in \Theta = (0, 1)$ is an unknown parameter, and $L > 0$ known. All limits are taken as $n \rightarrow \infty$.

1. Find the complete sufficient statistic for θ . Is the statistic minimal?

Solution: The Lebesgue density is

$$f(x; \theta) = \frac{d}{dx} F(x; \theta) = \frac{L^{1/\theta}}{\theta} x^{-(1+\theta)/\theta} \mathbb{1}(x > L) = h(x)c(\theta) \exp(T(x)w(\theta)),$$

where $h(x) = \mathbb{1}(x > L)$, $c(\theta) = L^{1/\theta}/\theta$, $T(x) = \log(x)$ and $w(\theta) = -(1 + \theta)/\theta$. Thus, $\mathcal{P}$ belongs to the exponential family.

By properties of the exponential family, $T(X_1, \dots, X_n) = \sum_{i=1}^n \log(X_i)$ is a sufficient statistic, and completeness follows from the fact that the parameter space

$$\left\{ - \left(1 + \frac{1}{\theta} \right) : \theta \in (0, 1) \right\} = (-\infty, -2)$$

contains an open set. Because the statistic is sufficient and complete, it follows by Bahadur theorem that is also minimal.

2. Find the method of moment estimator $\hat{\theta}_{\text{MM}}$ based on $\mathbb{E}_\theta[X_i]$.

Solution: The expected value is

$$\mathbb{E}_\theta[X_i] = \int x f(x; \theta) dx = \int_L^\infty \frac{L^{1/\theta}}{\theta} x^{-1/\theta} dx = -\frac{L^{1/\theta}}{1-\theta} x^{-(1-\theta)/\theta} \Big|_{x=L}^\infty = \frac{L}{1-\theta}.$$

Therefore, the methods of moment estimator is

$$\hat{\theta}_{\text{MM}} = \frac{\bar{X} - L}{\bar{X}}, \quad \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

3. Show that $\hat{\theta}_{\text{MM}}$ is downward biased but consistent for θ .

Solution: By Jensen's inequality (applied to the convex function $x \mapsto L/x$),

$$\mathbb{E}_\theta[\hat{\theta}_{\text{MM}}] = 1 - \mathbb{E}_\theta \left[\frac{L}{\bar{X}} \right] < 1 - \frac{L}{\mathbb{E}_\theta[\bar{X}]} = \theta.$$

This inequality is strict as $x \mapsto L/x$ is strictly convex and $\bar{X}$ is not a.s. constant. Hence $\hat{\theta}_{\text{MM}}$ underestimates θ on average. This can also be shown using the Cauchy-Schwarz inequality.

By the LLN, $\bar{X} \rightarrow_{\mathbb{P}} \mathbb{E}_{\theta}[X_i] = \frac{L}{1-\theta}$ because $\mathbb{E}_{\theta}[X_i^2] < \infty$. Thus, the CMT implies

$$\hat{\theta}_{\text{MM}} = \frac{\bar{X} - L}{\bar{X}} \rightarrow_{\mathbb{P}} \frac{\frac{L}{1-\theta} - L}{\frac{L}{1-\theta}} = \theta.$$

4. Find the maximum likelihood estimator $\hat{\theta}_{\text{ML}}$.

Solution: The log-likelihood function is

$$\ell_n(\theta) = -\frac{1+\theta}{\theta} \sum_{i=1}^n \log(X_i) - n \log(\theta) + \frac{n}{\theta} \log L,$$

and therefore

$$\ell_{\theta,n}(\theta) = \frac{d}{d\theta} \ell_n(\theta) = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^n \log(X_i) - \frac{n}{\theta^2} \log L = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^n \log(X_i/L),$$

and

$$\ell_{\theta\theta,n}(\theta) = \frac{d^2}{d\theta^2} \ell_n(\theta) = \frac{n}{\theta^2} - \frac{2}{\theta^3} \sum_{i=1}^n \log(X_i) + \frac{2n}{\theta^3} \log L = \frac{1}{\theta} \frac{d}{dx} \ell_n(\theta) - \frac{1}{\theta^3} \sum_{i=1}^n \log(X_i).$$

This implies that

$$\hat{\theta}_{\text{ML}} = \frac{1}{n} \sum_{i=1}^n \log\left(\frac{X_i}{L}\right)$$

because $\ell_{\theta,n}(\hat{\theta}_{\text{ML}}) = 0$, $\ell_{\theta\theta,n}(\hat{\theta}_{\text{ML}}) < 0$, and the boundary conditions also hold.

5. Show that $\hat{\theta}_{\text{ML}}$ is unbiased and consistent for θ .

Solution: Let $Y = \log(X/L)$. Then, $\mathbb{P}_{\theta}(Y \leq y) = \mathbb{P}_{\theta}(\log(X/L) \leq y) = \mathbb{P}_{\theta}(X \leq L \exp(y))$.

This implies that $\mathbb{P}_{\theta}(Y \leq y) = (1 - \exp(-y/\theta)) \mathbb{1}(y > 0)$ and the Lebesgue density is

$$f_Y(y; \theta) = \frac{1}{\theta} \exp\left(-\frac{y}{\theta}\right) \mathbb{1}(y > 0).$$

which means that $Y \sim \text{Exponential}(\theta)$.

From the properties of the exponential distribution, $\mathbb{E}_{\theta}[\hat{\theta}_{\text{ML}}] = \theta$, and therefore $\hat{\theta}_{\text{ML}}$ is unbiased for θ . Furthermore, $\hat{\theta}_{\text{ML}} \rightarrow_{\mathbb{P}} \theta$ because $\mathbb{E}_{\theta}[\log(X/L)^2] < \infty$ and the result follows by the LLN.

6. Find the UMVU estimator of θ . Is the estimator asymptotically efficient?

Solution: The estimator $\hat{\theta}_{\text{ML}}$ is based on complete (minimal) sufficient statistic and is unbiased. Thus, it is UMVUE. Furthermore, if an unbiased estimator minimizes variance for each $n \geq 1$, then it must be asymptotically efficient as well. Furthermore, if an unbiased estimator minimizes variance for each $n \geq 1$, then it must be asymptotically efficient as well.

7. Consider the following one-sided hypothesis testing problem:

$$H_0 : \theta \geq \theta_0 \quad \text{vs.} \quad H_1 : \theta < \theta_0, \quad \text{for some fixed } \theta_0 \in (0, 1).$$

(a) Find the uniformly most powerful (UMP) testing procedure of level $\alpha \in (0, 1)$.

Solution: we will use Karlin-Rubin, so we start by checking the MLR condition. Take $\theta_0 < \theta_1$, then

$$\begin{aligned} \frac{f(X, \theta_1)}{f(X, \theta_0)} &= \frac{L^{1/\theta_1}/\theta_1}{L^{1/\theta_0}/\theta_0} \cdot \frac{\exp\{-(1 + 1/\theta_1) \sum_{i=1}^n \log(X_i)\}}{\exp\{-(1 + 1/\theta_0) \sum_{i=1}^n \log(X_i)\}} \\ &= L^{-(1/\theta_0 - 1/\theta_1)} \frac{\theta_0}{\theta_1} \exp\left\{(1/\theta_0 - 1/\theta_1) \sum_{i=1}^n \log(X_i)\right\} \end{aligned}$$

which is increasing in $\sum_{i=1}^n \log(X_i)$, and hence in $n^{-1} \sum_{i=1}^n \log(X_i/L)$, so the MLR property is satisfied. Therefore, the UMP test rejects if

$$\frac{1}{n} \sum_{i=1}^n \log(X_i/L) < c \quad \Longleftrightarrow \quad \frac{1}{\theta_0 n} \sum_{i=1}^n \log(X_i/L) = \frac{\hat{\theta}_{\text{ML}}}{\theta_0} < c^*$$

where, by properties of the exponential distribution,

$$\frac{\hat{\theta}_{\text{ML}}}{\theta_0} \sim \text{Gamma}(n, 1/n)$$

which is pivotal. The constant c^* is

$$\mathbb{P}_{\theta_0} \left[\frac{\hat{\theta}_{\text{ML}}}{\theta_0} < c^* \right] = \alpha \quad \Longleftrightarrow \quad c^* = \gamma_{n, 1/n}(\alpha)$$

where $\gamma_{n, 1/n}(\alpha)$ is the α -quantile from a $\text{Gamma}(n, 1/n)$. (Recall that if $W_1, \dots, W_n$ are i.i.d. $\text{Exponential}(\lambda)$, then $\sum_{i=1}^n W_i \sim \text{Gamma}(n, \lambda)$.)

(b) Find the power function $\beta_n(\theta)$ of the UMP test and sketch its graph.

Solution: The power function is

$$\beta_n(\theta) = \mathbb{P}_{\theta} \left[\frac{\hat{\theta}_{\text{ML}}}{\theta_0} < \gamma_{n, 1/n}(\alpha) \right] = \mathbb{P}_{\theta} \left[\frac{\hat{\theta}_{\text{ML}}}{\theta} < \frac{\theta_0}{\theta} \gamma_{n, 1/n}(\alpha) \right] = \Gamma_{n, 1/n} \left(\frac{\theta_0}{\theta} \gamma_{n, 1/n}(\alpha) \right),$$

$\theta \in (0, 1)$, where $\Gamma_{n, 1/n}(\cdot)$ denotes the c.d.f. of a $\text{Gamma}(n, 1/n)$ random variable. The function is plotted in Figure 1. It is easy to see that $\beta_n(0) = 1$ and β_n is decreasing and $\beta_n(\theta_0) = \alpha$.

(c) Show that the UMP test is consistent.

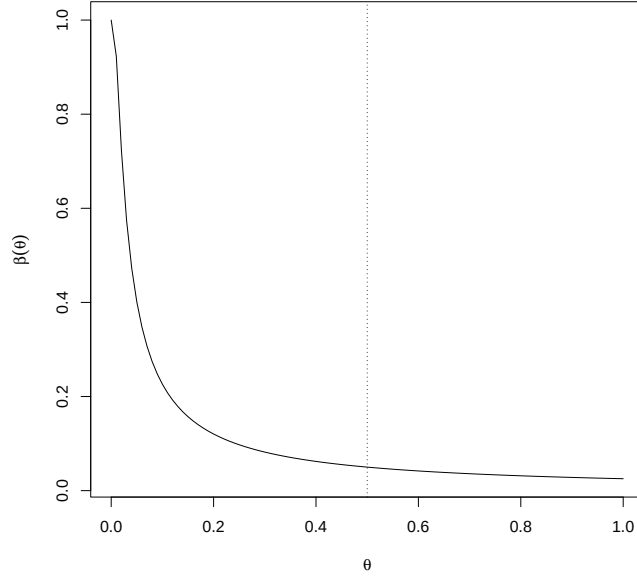


Figure 1: Power function for $L = 2$, $\theta_0 = 0.5$, $n = 10$, $\alpha = 0.05$.

Solution: Let $Y_n \sim \text{Gamma}(n, 1/n)$. By the CLT,

$$\begin{aligned} \alpha &= \mathbb{P}(Y_n \leq \gamma_{n,1/n}(\alpha)) \\ &= \mathbb{P}(\sqrt{n}(Y_n - 1) \leq \sqrt{n}(\gamma_{n,1/n}(\alpha) - 1)) \\ &\rightarrow \Phi\left(\lim_{n \rightarrow \infty} \sqrt{n}(\gamma_{n,1/n}(\alpha) - 1)\right), \end{aligned}$$

which implies that $\sqrt{n}(\gamma_{n,1/n}(\alpha) - 1) \rightarrow z_\alpha$, the α -quantile of a standard normal. Therefore,

$$\begin{aligned} \beta_n(\theta) &= \Gamma_{n,1/n}\left(\frac{\theta_0}{\theta}\gamma_{n,1/n}(\alpha)\right) \\ &= \mathbb{P}\left(Y_n \leq \frac{\theta_0}{\theta}\gamma_{n,1/n}(\alpha)\right) \\ &= \mathbb{P}\left(\sqrt{n}(Y_n - 1) \leq \sqrt{n}\left(\frac{\theta_0}{\theta}\gamma_{n,1/n}(\alpha) - 1\right)\right) \\ &= \mathbb{P}\left(\sqrt{n}(Y_n - 1) \leq \sqrt{n}\frac{\theta_0}{\theta}(\gamma_{n,1/n}(\alpha) - 1) + \sqrt{n}\left(\frac{\theta_0}{\theta} - 1\right)\right) \\ &\rightarrow \Phi\left(\frac{\theta_0}{\theta}z_\alpha + \lim_{n \rightarrow \infty} \sqrt{n}\left(\frac{\theta_0}{\theta} - 1\right)\right) \\ &= \begin{cases} \Phi(z_\alpha) = \alpha & \theta = \theta_0 \\ \Phi(\infty) = 1 & \theta < \theta_0, \\ \Phi(-\infty) = 0 & \theta > \theta_0 \end{cases} \end{aligned}$$

which implies that the UMP test is consistent.

- (d) Construct a finite sample exact $(1 - \alpha)$ confidence interval for θ .

Solution: we have

$$1 - \alpha = \mathbb{P}_\theta \left[\frac{\hat{\theta}_{\text{ML}}}{\theta} \geq \gamma_{n,1/n} \right] = \mathbb{P}_\theta \left[\theta \leq \frac{\hat{\theta}_{\text{ML}}}{\gamma_{n,1/n}} \right]$$

so the confidence interval is

$$\text{CI}_{n,1-\alpha}(\theta) = \left[0, \frac{\hat{\theta}_{\text{ML}}}{\gamma_{n,1/n}} \right]$$

2 Minimal Sufficient Statistics (25 points)

Let a random sample $\mathbf{X}$ with values in the sample space $\mathcal{X}$ have a distribution $\mathbb{P}_\theta$, where $\mathcal{P} = \{\mathbb{P}_\theta : \theta \in \Theta\}$ is a parametric family dominated by some (σ -finite) measure with strictly positive densities $\{f(\mathbf{x}; \theta) : \theta \in \Theta\}$. To avoid measurability issues, assume $\mathcal{X}$ is finite or countable and all its subsets are measurable. Suppose that there exists a statistic T such that, for every $\mathbf{x}, \mathbf{y} \in \mathcal{X}$, the ratio $f(\mathbf{x}; \theta)/f(\mathbf{y}; \theta)$ does not depend on θ if and only if $T(\mathbf{x}) = T(\mathbf{y})$.

1. Let $T^{-1}(t) = \{\mathbf{x} : T(\mathbf{x}) = t\}$ for each $t \in R_T = \{t : t = T(\mathbf{x}), \mathbf{x} \in \mathcal{X}\}$.

Show that there is a function $R_T \ni t \mapsto \mathbf{y}_t \in \mathcal{X}$ such that for all $\mathbf{x} \in \mathcal{X}$ we have

$$T(\mathbf{x}) = T(\mathbf{y}_{T(\mathbf{x})}).$$

Solution: Choose an arbitrary representative $\mathbf{y}_t \in T^{-1}(t)$. Then, for any $\mathbf{x} \in \mathcal{X}$ we have $\mathbf{x} \in T^{-1}(T(\mathbf{x}))$ and $\mathbf{y}_{T(\mathbf{x})} \in T^{-1}(T(\mathbf{x}))$. Therefore, $T(\mathbf{x}) = T(\mathbf{y}_{T(\mathbf{x})})$.

2. Show that T is a sufficient statistic (for $\mathcal{P}$).

Solution: Since T maps $\mathbf{x}$ and $\mathbf{y}_{T(\mathbf{x})}$ to the same value, the ratio $f(\mathbf{x}; \theta)/f(\mathbf{y}_{T(\mathbf{x})}; \theta)$ does not depend on θ , so it is a function only of $\mathbf{x}$. Denote it $h(\mathbf{x})$. We have

$$f(\mathbf{x}; \theta) = h(\mathbf{x}) \underbrace{f(\mathbf{y}_{T(\mathbf{x})}; \theta)}_{g(T(\mathbf{x}); \theta)},$$

and thus T is a sufficient statistic by the factorization theorem.

3. Let S be another sufficient statistic (for $\mathcal{P}$). Show that, for every $\mathbf{x}, \mathbf{y} \in \mathcal{X}$, the equality $S(\mathbf{x}) = S(\mathbf{y})$ implies that the ratio $f(\mathbf{x}; \theta)/f(\mathbf{y}; \theta)$ does not depend on θ .

Solution: By the factorization theorem, this ratio is

$$\frac{f(\mathbf{x}; \theta)}{f(\mathbf{y}; \theta)} = \frac{h(\mathbf{x})g(S(\mathbf{x}); \theta)}{h(\mathbf{y})g(S(\mathbf{y}); \theta)} = \frac{h(\mathbf{x})}{h(\mathbf{y})},$$

where the second equality uses $S(\mathbf{x}) = S(\mathbf{y})$.

4. Show that T is a minimal sufficient statistic (for $\mathcal{P}$).

Hint: $A(\cdot)$ is a function of $B(\cdot)$ if and only if $B(u) = B(v) \implies A(u) = A(v)$ for all u, v .

Solution: Using the previous part,

$$S(\mathbf{x}) = S(\mathbf{y}) \implies T(\mathbf{x}) = T(\mathbf{y})$$

for any sufficient statistic S and for every $\mathbf{x}, \mathbf{y} \in \mathcal{X}$. As a result, using the hint, T is a minimal sufficient statistic.

5. Suppose that $\mathcal{P}$ belongs to the two-dimensional exponential family.
Give sufficient conditions for the existence of a two-dimensional minimal sufficient statistics.

Solution: We have

$$\begin{aligned}\frac{f(\mathbf{x}; \theta)}{f(\mathbf{y}; \theta)} &= \frac{c(\theta)h(\mathbf{x}) \exp(T_1(\mathbf{x})w_1(\theta) + T_2(\mathbf{x})w_2(\theta))}{c(\theta)h(\mathbf{y}) \exp(T_1(\mathbf{y})w_1(\theta) + T_2(\mathbf{y})w_2(\theta))} \\ &= \frac{h(\mathbf{x})}{h(\mathbf{y})} \exp([T_1(\mathbf{x}) - T_1(\mathbf{y})]w_1(\theta) + [T_2(\mathbf{x}) - T_2(\mathbf{y})]w_2(\theta)).\end{aligned}$$

Obviously, the equality $(T_1(\mathbf{x}), T_2(\mathbf{x})) = (T_1(\mathbf{y}), T_2(\mathbf{y}))$ implies that this ratio does not depend on θ . For the other direction, $(w_1(\theta) - w_1(\theta_0), w_2(\theta) - w_2(\theta_0))$ need to be linearly independent as functions of $\theta \in \Theta$ for any fixed θ_0 .

3 MSE Prediction and Identification (25 points)

Let $(Y, \mathbf{X}, \mathbf{W})$ be a random element, where Y has values in $\mathbb{R}$, $\mathbf{X}$ has values in $\mathbb{R}^{d_x}$, and $\mathbf{W}$ has values in $\mathbb{R}^{d_w}$. Assume $\mathbb{E}[Y^2] < \infty$ and $\mathbb{E}[\mathbf{X}'\mathbf{X}] < \infty$. Consider the following MSE prediction problem with respect to the class of partially linear models:

$$(\beta^*, h^*) \in \arg \min_{g \in \mathcal{G}_{\text{PL}}} \mathbb{E}[(Y - g(\mathbf{X}, \mathbf{W}))^2]$$

where the feasibility set is

$$\mathcal{G}_{\text{PL}} = \{g(\mathbf{x}, \mathbf{w}) = \mathbf{x}'\beta + h(\mathbf{w}) : \beta \in \mathbb{R}^{d_x}, h \in \mathcal{H}\}, \quad \mathcal{H} = \{h : \mathbb{E}[|h(\mathbf{W})|^2] < \infty\}.$$

1. Suppose that β^* is known, and find the ($\mathbb{P}_{\mathbf{W}}$ -a.s. unique) solution h^* .

Solution: Given β^* , we have

$$\min_{g \in \mathcal{G}_{\text{PL}}} \mathbb{E}[(Y - g(\mathbf{X}, \mathbf{W}))^2] = \min_{h \in \mathcal{H}} \mathbb{E}[((Y - \mathbf{X}'\beta^*) - h(\mathbf{W}))^2]$$

which implies that

$$h^*(\mathbf{W}) = \mathbb{E}[Y - \mathbf{X}'\beta^* | \mathbf{W}] = \mathbb{E}[Y | \mathbf{W}] - \mathbb{E}[\mathbf{X} | \mathbf{W}]'\beta^*.$$

2. Given the solution in the previous part, find the set of optimal solutions for the parameter β .

Solution: Given $h^*(\mathbf{W}; \beta) = \mathbb{E}[Y | \mathbf{W}] - \mathbb{E}[\mathbf{X} | \mathbf{W}]'\beta$, we have

$$\min_{g \in \mathcal{G}_{\text{PL}}} \mathbb{E}[(Y - g(\mathbf{X}, \mathbf{W}))^2] = \min_{\beta \in \mathbb{R}^{d_x}} \mathbb{E}[((Y - \mathbb{E}[Y | \mathbf{W}]) - (\mathbf{X} - \mathbb{E}[\mathbf{X} | \mathbf{W}])'\beta)^2]$$

which implies that β is the best linear predictor and thus solves the (normal) equations

$$\mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X} | \mathbf{W}])(\mathbf{X} - \mathbb{E}[\mathbf{X} | \mathbf{W}])'\beta] = \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X} | \mathbf{W}])(Y - \mathbb{E}[Y | \mathbf{W}])],$$

or, equivalently, $\mathbb{E}[\mathbb{V}[\mathbf{X} | \mathbf{W}]]\beta = \mathbb{E}[\text{Cov}[\mathbf{X}, Y | \mathbf{W}]]$. For example,

$$\beta^* = \mathbb{E}[\mathbb{V}[\mathbf{X} | \mathbf{W}]]^- \mathbb{E}[\text{Cov}[\mathbf{X}, Y | \mathbf{W}]],$$

where $\mathbf{A}^-$ denotes a generalized inverse. The solution β^* is unique if $\mathbb{E}[\mathbb{V}[\mathbf{X} | \mathbf{W}]] > 0$ (meaning that the matrix is positive definite).

3. Give sufficient conditions so that $(\beta^*, h^*(\mathbf{W}))$ are (a.s.) unique.

Solution: From the derivation above, $\mathbb{E}[\mathbb{V}[\mathbf{X} | \mathbf{W}]] > 0$ is sufficient.

4. Assume that $\mathbb{E}[Y | \mathbf{X}, \mathbf{W}] = \mathbf{X}'\beta_0 + h_0(\mathbf{W})$, with $(\beta_0, h_0(\mathbf{W}))$ are (a.s.) unique.

(a) Show that $\beta^* = \beta_0$ and $h^*(\mathbf{W}) = h_0(\mathbf{W})$ (a.s.).

Solution: We have

$$\begin{aligned}
& \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])(Y - \mathbb{E}[Y|\mathbf{W}])] \\
&= \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])(\mathbb{E}[Y|\mathbf{X}, \mathbf{W}] - \mathbb{E}[Y|\mathbf{W}])] \\
&= \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])(\mathbf{X}'\beta_0 + h_0(\mathbf{W}) - \mathbb{E}[Y|\mathbf{W}])] \\
&= \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])\mathbf{X}']\beta_0 + \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])(h_0(\mathbf{W}) - \mathbb{E}[Y|\mathbf{W}])] \\
&= \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])']\beta_0
\end{aligned}$$

and therefore

$$\beta^* = (\mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])'])^{-1} \mathbb{E}[(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])(Y - \mathbb{E}[Y|\mathbf{W}])] = \beta_0.$$

Finally,

$$\begin{aligned}
h^*(\mathbf{W}) &= \mathbb{E}[Y|\mathbf{W}] - \mathbb{E}[\mathbf{X}|\mathbf{W}]'\beta^* \\
&= \mathbb{E}[\mathbb{E}[Y|\mathbf{X}, \mathbf{W}]|\mathbf{W}] - \mathbb{E}[\mathbf{X}|\mathbf{W}]'\beta_0 \\
&= \mathbb{E}[\mathbf{X}'\beta_0 + h_0(\mathbf{W})|\mathbf{W}] - \mathbb{E}[\mathbf{X}|\mathbf{W}]'\beta_0 \\
&= h_0(\mathbf{W}).
\end{aligned}$$

(b) Show that

$$\min_{h \in \mathcal{H}} \mathbb{E}[(Y - h(\mathbf{W}))^2] - \min_{g \in \mathcal{G}_{\text{PL}}} \mathbb{E}[(Y - g(\mathbf{X}, \mathbf{W}))^2] = \beta_0' \mathbb{E}[\mathbb{V}[\mathbf{X}|\mathbf{W}]]\beta_0.$$

When is it strictly better (in MSE sense) to include $\mathbf{X}$ when constructing the best MSE predictor of Y given $\mathbf{X}$ and $\mathbf{W}$?

Solution: We have

$$\begin{aligned}
& \mathbb{E}[(Y - \mathbf{X}'\beta^* - h^*(\mathbf{W}))^2] \\
&= \mathbb{E}[(Y - \mathbb{E}[Y|\mathbf{W}] - (\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])'\beta^*)^2] \\
&= \mathbb{E}[(Y - \mathbb{E}[Y|\mathbf{W}])^2] - 2\mathbb{E}[(Y - \mathbb{E}[Y|\mathbf{W}])(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])'\beta^*] + \mathbb{E}[((\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])'\beta^*)^2] \\
&= \mathbb{E}[\mathbb{V}[Y|\mathbf{W}]] - \mathbb{E}[((\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])'\beta_0)^2] \\
&= \mathbb{E}[\mathbb{V}[Y|\mathbf{W}]] - \beta_0' \mathbb{E}[\mathbb{V}[\mathbf{X}|\mathbf{W}]]\beta_0
\end{aligned}$$

because

$$\begin{aligned}
\mathbb{E}[(Y - \mathbb{E}[Y|\mathbf{W}])(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])'\beta^*] &= \mathbb{E}[(\mathbb{E}[Y|\mathbf{X}, \mathbf{W}] - \mathbb{E}[Y|\mathbf{W}])(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])'\beta^*] \\
&= \mathbb{E}[(\mathbf{X}'\beta_0 - \mathbb{E}[\mathbf{X}|\mathbf{W}]'\beta_0)(\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])'\beta_0] \\
&= \mathbb{E}[((\mathbf{X} - \mathbb{E}[\mathbf{X}|\mathbf{W}])'\beta_0)^2].
\end{aligned}$$

The result follows because $\min_{h \in \mathcal{H}} \mathbb{E}[(Y - h(\mathbf{W}))^2] = \mathbb{E}[\mathbb{V}[Y|\mathbf{W}]]$.

Finally, if $\beta_0' \mathbb{E}[\mathbb{V}[\mathbf{X}|\mathbf{W}]]\beta_0 > 0$, then the best MSE predictor $\mathbb{E}[Y|\mathbf{X}]$ has higher MSE than the best MSE predictor $\mathbb{E}[Y|\mathbf{X}, \mathbf{W}]$. In words, including the additional covariates $\mathbf{X}$ reduces MSE prediction when they are not irrelevant (i.e., $\beta_0 \neq \mathbf{0}$). The result is true in more generality than just for the partially linear model under consideration.

Table 1.1. Discrete Distributions on $\mathcal{R}$

Uniform	p.d.f.	$1/m, x = a_1, \dots, a_m$
	m.g.f.	$\sum_{j=1}^m e^{a_j t}/m, t \in \mathcal{R}$
$DU(a_1, \dots, a_m)$	Expectation	$\sum_{j=1}^m a_j/m$
	Variance	$\sum_{j=1}^m (a_j - \bar{a})^2/m, \bar{a} = \sum_{j=1}^m a_j/m$
	Parameter	$a_i \in \mathcal{R}, m = 1, 2, \dots$
Binomial	p.d.f.	$\binom{n}{x} p^x (1-p)^{n-x}, x = 0, 1, \dots, n$
	m.g.f.	$(pe^t + 1 - p)^n, t \in \mathcal{R}$
$Bi(p, n)$	Expectation	np
	Variance	$np(1-p)$
	Parameter	$p \in [0, 1], n = 1, 2, \dots$
Poisson	p.d.f.	$\theta^x e^{-\theta}/x!, x = 0, 1, 2, \dots$
	m.g.f.	$e^{\theta(e^t - 1)}, t \in \mathcal{R}$
$P(\theta)$	Expectation	θ
	Variance	θ
	Parameter	$\theta > 0$
Geometric	p.d.f.	$(1-p)^{x-1}p, x = 1, 2, \dots$
	m.g.f.	$pe^t/[1 - (1-p)e^t], t < -\log(1-p)$
$G(p)$	Expectation	$1/p$
	Variance	$(1-p)/p^2$
	Parameter	$p \in [0, 1]$
Hyper-geometric	p.d.f.	$\binom{n}{x} \binom{m}{r-x} / \binom{N}{r}$
		$x = 0, 1, \dots, \min\{r, n\}, r - x \leq m$
$HG(r, n, m)$	m.g.f.	No explicit form
	Expectation	rn/N
	Variance	$rn m(N-r)/[N^2(N-1)]$
	Parameter	$r, n, m = 1, 2, \dots, N = n + m$
Negative binomial	p.d.f.	$\binom{x-1}{r-1} p^r (1-p)^{x-r}, x = r, r+1, \dots$
	m.g.f.	$p^r e^{rt}/[1 - (1-p)e^t]^r, t < -\log(1-p)$
$NB(p, r)$	Expectation	r/p
	Variance	$r(1-p)/p^2$
	Parameter	$p \in [0, 1], r = 1, 2, \dots$
Log-distribution	p.d.f.	$-(\log p)^{-1} x^{-1} (1-p)^x, x = 1, 2, \dots$
	m.g.f.	$\log[1 - (1-p)e^t]/\log p, t \in \mathcal{R}$
$L(p)$	Expectation	$-(1-p)/(p \log p)$
	Variance	$-(1-p)[1 + (1-p)/\log p]/(p^2 \log p)$
	Parameter	$p \in (0, 1)$

All p.d.f.'s are w.r.t. counting measure.

Table 1.2. Distributions on $\mathcal{R}$ with Lebesgue p.d.f.'s

Uniform	p.d.f.	$(b-a)^{-1}I_{(a,b)}(x)$
	m.g.f.	$(e^{bt} - e^{at})/[(b-a)t], \quad t \in \mathcal{R}$
$U(a, b)$	Expectation	$(a+b)/2$
	Variance	$(b-a)^2/12$
	Parameter	$a, b \in \mathcal{R}, \quad a < b$
Normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}e^{-(x-\mu)^2/2\sigma^2}$
	m.g.f.	$e^{\mu t + \sigma^2 t^2/2}, \quad t \in \mathcal{R}$
$N(\mu, \sigma^2)$	Expectation	μ
	Variance	σ^2
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$
Exponential	p.d.f.	$\theta^{-1}e^{-(x-a)/\theta}I_{(a,\infty)}(x)$
	m.g.f.	$e^{at}(1-\theta t)^{-1}, \quad t < \theta^{-1}$
$E(a, \theta)$	Expectation	$\theta + a$
	Variance	θ^2
	Parameter	$\theta > 0, \quad a \in \mathcal{R}$
Chi-square	p.d.f.	$\frac{1}{\Gamma(k/2)2^{k/2}}x^{k/2-1}e^{-x/2}I_{(0,\infty)}(x)$
	m.g.f.	$(1-2t)^{-k/2}, \quad t < 1/2$
χ_k^2	Expectation	k
	Variance	$2k$
	Parameter	$k = 1, 2, \dots$
Gamma	p.d.f.	$\frac{1}{\Gamma(\alpha)\gamma^\alpha}x^{\alpha-1}e^{-x/\gamma}I_{(0,\infty)}(x)$
	m.g.f.	$(1-\gamma t)^{-\alpha}, \quad t < \gamma^{-1}$
$\Gamma(\alpha, \gamma)$	Expectation	$\alpha\gamma$
	Variance	$\alpha\gamma^2$
	Parameter	$\gamma > 0, \quad \alpha > 0$
Beta	p.d.f.	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}x^{\alpha-1}(1-x)^{\beta-1}I_{(0,1)}(x)$
	m.g.f.	No explicit form
$B(\alpha, \beta)$	Expectation	$\alpha/(\alpha+\beta)$
	Variance	$\alpha\beta/[(\alpha+\beta+1)(\alpha+\beta)^2]$
	Parameter	$\alpha > 0, \quad \beta > 0$
Cauchy	p.d.f.	$\frac{1}{\pi\sigma}\left[1+\left(\frac{x-\mu}{\sigma}\right)^2\right]^{-1}$
	ch.f.	$e^{\sqrt{-1}\mu t - \sigma t }$
$C(\mu, \sigma)$	Expectation	Does not exist
	Variance	Does not exist
	Parameter	$\mu \in \mathcal{R}, \quad \sigma > 0$

Table 1.2. (continued)

t-distribution	p.d.f.	$\frac{\Gamma[(n+1)/2]}{\sqrt{n\pi}\Gamma(n/2)} \left(1 + \frac{x^2}{n}\right)^{-(n+1)/2}$
	ch.f.	No explicit form
t_n	Expectation	0, ($n > 1$)
	Variance	$n/(n-2)$, ($n > 2$)
	Parameter	$n = 1, 2, \dots$
F-distribution	p.d.f.	$\frac{n^{n/2}m^{m/2}\Gamma[(n+m)/2]x^{n/2-1}}{\Gamma(n/2)\Gamma(m/2)(m+nx)^{(n+m)/2}}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$F_{n,m}$	Expectation	$m/(m-2)$, ($m > 2$)
	Variance	$2m^2(n+m-2)/[n(m-2)^2(m-4)]$, ($m > 4$)
	Parameter	$n = 1, 2, \dots$, $m = 1, 2, \dots$
Log-normal	p.d.f.	$\frac{1}{\sqrt{2\pi}\sigma}x^{-1}e^{-(\log x - \mu)^2/2\sigma^2}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$LN(\mu, \sigma^2)$	Expectation	$e^{\mu+\sigma^2/2}$
	Variance	$e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$
Weibull	p.d.f.	$\frac{\alpha}{\theta}x^{\alpha-1}e^{-x^\alpha/\theta}I_{(0,\infty)}(x)$
	ch.f.	No explicit form
$W(\alpha, \theta)$	Expectation	$\theta^{1/\alpha}\Gamma(\alpha^{-1} + 1)$
	Variance	$\theta^{2/\alpha} \left\{ \Gamma(2\alpha^{-1} + 1) - [\Gamma(\alpha^{-1} + 1)]^2 \right\}$
	Parameter	$\theta > 0$, $\alpha > 0$
Double Exponential	p.d.f.	$\frac{1}{2\theta}e^{- x-\mu /\theta}$
	m.g.f.	$e^{\mu t}/(1 - \theta^2 t^2)$, $ t < \theta^{-1}$
$DE(\mu, \theta)$	Expectation	μ
	Variance	$2\theta^2$
	Parameter	$\mu \in \mathcal{R}$, $\theta > 0$
Pareto	p.d.f.	$\theta a^\theta x^{-(\theta+1)}I_{(a,\infty)}(x)$
	ch.f.	No explicit form
$Pa(a, \theta)$	Expectation	$\theta a/(\theta - 1)$, ($\theta > 1$)
	Variance	$\theta a^2/[(\theta - 1)^2(\theta - 2)]$, ($\theta > 2$)
	Parameter	$\theta > 0$, $a > 0$
Logistic	p.d.f.	$\sigma^{-1}e^{-(x-\mu)/\sigma}/[1 + e^{-(x-\mu)/\sigma}]^2$
	m.g.f.	$e^{\mu t}\Gamma(1 + \sigma t)\Gamma(1 - \sigma t)$, $ t < \sigma^{-1}$
$LG(\mu, \sigma)$	Expectation	μ
	Variance	$\sigma^2\pi^2/3$
	Parameter	$\mu \in \mathcal{R}$, $\sigma > 0$